MH2000 : Experimental Methods
Thermodynamic Measurements : Lecture 2
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Why measure Gibbs energies?

In Lecture 1 we imposed known chemical potentials, for example with CO–CO_2 or H_2–H_2O gas mixtures, and observed how a sample responded. Lecture 2 asks the reverse question: how do we turn measurements into the standard Gibbs energy, enthalpy and entropy of a reaction? These are the numbers stored in thermodynamic databases and used in process calculations, from the deoxidation of steel to the refining of silicon.

A practical question: How do we know that Cu_2O is stable at 1000 K, and how stable it is? Someone had to measure it, and there are only three basic routes to that number.

One equation, three routes

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ = -RT \ln K = -zFE^\circ$$

ΔG°, ΔH°, ΔS°: standard Gibbs energy and enthalpy (J mol^{-1}) and entropy (J mol^{-1} K^{-1}) of reaction;
K: equilibrium constant, defined with activities (dimensionless);
z: number of electrons transferred per reaction as written;
F: Faraday constant, 96 485 C mol^{-1};
E°: standard cell voltage (V).

Each part of the lecture uses a different piece of this equation. We can calculate ΔG° from tabulated data (Gibbs energy function), extract ΔH° and ΔS° from equilibrium constants measured at several temperatures (second- and third-law methods), or read ΔG° directly as a voltage (electrochemical cells).

The Gibbs energy function (gef)

Fitted expressions such as $\Delta G^\circ = A + BT$ are convenient but approximate, and their error grows outside the temperature range they were fitted to. Integrating heat-capacity data for every reaction is exact but tedious. The Gibbs energy function combines the precision of the tables with the ease of a formula. For each substance, thermodynamic tables such as NIST-JANAF list

$$\mathbf{gef} = -(G^\circ_T - H^\circ_{298}) / T$$

This quantity changes slowly with temperature, so a table with entries every 100 K can be interpolated accurately. For a reaction, gef values are combined like any other property, products minus reactants, and

$$\Delta G^\circ_T = \Delta H^\circ_{298} - T\cdot\Delta\mathrm{gef}$$

Example: For $4Cu(s) + O_2(g) = 2Cu_2O(s)$ at 1000 K, the tables give Δgef = 2(127.769) − 4(46.261) − 220.875 = −150.381 J K^{-1} and ΔH°_{298} = 2(−167.688) = −335.376 kJ. Hence ΔG°_{1000} = −335.376 + 150.381 ≈ −185.0 kJ, close to the −185.9 kJ obtained from a simple A + BT expression.

Sign convention: the tables list gef with the minus sign so that the values are positive. The lecture appendix writes the same equations with $\Delta[(G^\circ_T - H^\circ_{298})/T]$, which flips the sign of that term. Both are correct.

Second-law method: equilibrium constants at several temperatures

Combining $\Delta G^\circ = -RT \ln K$ with $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$ and dividing by −RT gives the van 't Hoff form:

$$\ln K = -(\Delta H^\circ/R)\cdot(1/T) + \Delta S^\circ/R$$

A plot of ln K against 1/T is a straight line: the slope gives $-\Delta H^\circ/R$ and the intercept gives $\Delta S^\circ/R$. The results are mean values over the temperature range studied, because ΔH° and ΔS° are assumed constant.

For the vaporization Cu(l) = Cu(g), measured copper vapor pressures give ln K = $-3.679\times10^4/T + 24.47$ (p_{Cu} in Pa), so $\Delta H° = 305.9$ kJ mol^{-1} and $\Delta S° = 203.4$ J mol^{-1} K^{-1} at a mean temperature of about 1747 K. Correcting to 298 K with heat-capacity data gives $\Delta_{vap}H°_{298} = 330.7$ kJ mol^{-1}.

The second-law sigma method follows the same idea but allows the heat capacities to vary with temperature. The measured ln K values and the heat-capacity terms are combined into one quantity, Σ, which is again linear in 1/T. The result is a complete multi-term expression, $\Delta G° = A + BT \ln T + CT^2 + D/T + ET$.

Third-law method: one point is enough

$$\mathbf{\Delta H°_{298} = -RT \ln K + T\cdot\Delta gef}$$

With gef tables, every single measured equilibrium constant gives its own value of $\Delta H°_{298}$. If the experiment is sound, these values do not depend on the temperature at which they were measured. A systematic trend with temperature reveals an error in the temperature measurement or in K. This built-in check is why the third-law method is generally considered superior to the second-law methods.

Why it matters: a slope fitted over a narrow temperature range is very sensitive to scatter in the data. The third-law method treats each point independently, so it tests the data as well as evaluating them.

Which method when?

Method	You need	You get	Best used when
Second law	K at several temperatures	Mean ΔH° and ΔS° over the range	Data are sparse
Sigma method	K at several temperatures + c_p data	Full ΔG°(T) expression	Good heat-capacity data exist
Third law	K (even one point) + gef tables	$\Delta H°_{298}$ from every point	gef tables exist; best check for systematic errors

Electrochemical measurements: Gibbs energy as a voltage

In a reaction such as AO + B = BO + A, electrons move from B to A. In an electrochemical cell the two halves are separated: electrons must travel through an external wire and ions through an electrolyte. The voltage measured at zero current is the maximum electrical work per unit charge, which is set by the driving force of the reaction:

$$\mathbf{\Delta G = -zFE} \qquad \mathbf{E = E° - (RT/zF) \ln Q}$$

A positive cell voltage means a negative ΔG and a spontaneous reaction. The Nernst equation shows that composition changes the voltage, so voltage measurements give activities. When one electrode contains the pure component (a = 1) and the other the alloy or slag, $a_i = \exp(-zFE/RT)$. For example, z = 2 and E = 0.050 V at 1473 K give $a_{PbO} \approx 0.45$.

A useful cell needs an electrolyte that conducts only ions, one reversible reaction at each electrode, and materials that remain stable at high temperature without gas leakage or side reactions. Zirconia stabilized with CaO or Y_2O_3, which conducts O^{2-} ions, is the classic solid electrolyte.

Cell type	Electrodes	What it measures	Example
Formation cell	Different, e.g. metal + oxide against O_2 gas	ΔG°_f of an oxide as a function of T	Pb(l) | PbO(l) | O_2(1 atm), Pt; at 1100 K, $E^\circ \approx 0.54$ V
Concentration cell	Same substance at two different activities	Activity of a component	Pb–PbO cell with PbO dissolved in a PbO–SiO_2 slag

In industry: the oxygen probe, Cr–Cr_2O_3(s) | ZrO_2(CaO) | O (in Fe), is the same kind of cell on a stick. The Cr–Cr_2O_3 reference fixes a known oxygen potential, and the measured voltage gives the dissolved oxygen activity in liquid steel within seconds.

Before the exam: read Fruehan, Martonik and Turkdogan (1969) and be prepared to explain how an oxygen probe works.

Common pitfalls

1. E must be measured at zero current. Once current flows, the voltage drops and no longer equals $-\Delta G/zF$.
2. z counts electrons per reaction as written. Doubling the reaction doubles z and ΔG; E stays the same.
3. Check the standard state inside ln K. In the copper example $K = p_{Cu}$ in Pa, so ΔS° refers to a 1 Pa standard state.
4. ln and log differ by a factor of 2.303. Literature equations often use log.

Three questions to discuss

1. Why is the Gibbs energy function easier to tabulate and interpolate than G° itself?
2. A plot of ln K against 1/T has a slope of -2.0×10^4 K. What is ΔH°, and why is it only a mean value?
3. Third-law values of ΔH°_{298} rise steadily with temperature. What does this tell you about the experiment?

All three routes lead to the same quantity, ΔG°. Tables give it precisely, equilibrium constants give ΔH° and ΔS°, and electrochemical cells read it directly as a voltage. The third-law method and cell measurements also let us test the quality of the data.

References

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